Day - 1 [21/09/26].

Excel

Introduction

⇒ what is Excel ?

Microsoft Excel is a powerful spreadsheet tool used to store, organize, analyze and visualize data in analytics It acts as a beginner friendly yet powerful for Handling real world datasets.

⇒ why Excel

→ Easy to learn and use
→ Handles structured data efficiently
→ Powerful visualization tool
→ No coding required.

⇒ key components in Excel.

→ Workbook and worksheets.
→ Rows & columns.
R → Num (1, 2, 3 - - -)
C → alph (A, B, C - - -)
→ Cells → Intersection of Rows & columns.

⇒ Types of Data.

Text → names, categories
numbers → sales, quantity.
Dates → Time based analysis.
Boolean → True (1) / false (0)
Geographical data.

Important Excel Features

- Data extraction
- Data cleaning / transformation
  - removes duplicates
  - Handling the missing values
  - find and replace
  - Remove the irrelevant data and adding the necessary data.
- Formulas & functions
  - Basic (sum, average, count).
  - logical (if, and, or, not).
  - lookup (xlookup, vlookup, hlookup).
  - text (contact, upper, lower).
- Data visualization
  - charts
  - conditional formatting
  - sparklines
- pivot tables
  - summarize tables
  - Grap and filter data
  - Generate insights quickly.
- sorting & filtering.
- Data validation (Extracting the proper data).

Advantages :-

- Beginner friendly.
- widely used in companies.
- Integrates with tools like power BI.

Disadvantages :-

- For Large datasets not suitable.
- Mannual errors are possible.
- limited automation compared to coding too